

Unit 2 : Cloud Computing Architecture

Cloud computing architecture is like a virtual foundation for the internet that helps businesses and individuals access and use computing resources, like servers, storage, databases, and software, over the internet. It's like renting or sharing these resources instead of owning and managing them on your own physical computers.

2.2.2 Private Clouds

Advantages



ADVANTAGES



1) MORE CONTROL

Comprehensive control over dedicated hardware and resources, accessible only to authorized users.



2) SECURITY & PRIVACY

Enhanced security level compared to public models, through dedicated infrastructure and strict access management.



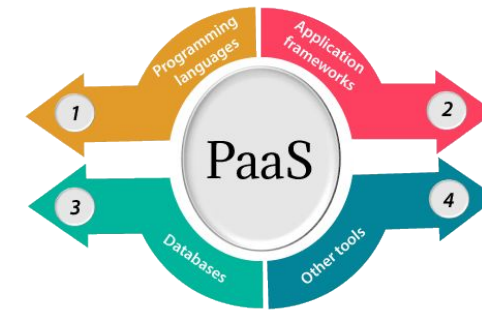
3) IMPROVED PERFORMANCE

Consistent, reliable performance with optimized speed and capacity tailored to organizational needs.

2.1 Cloud Reference Model

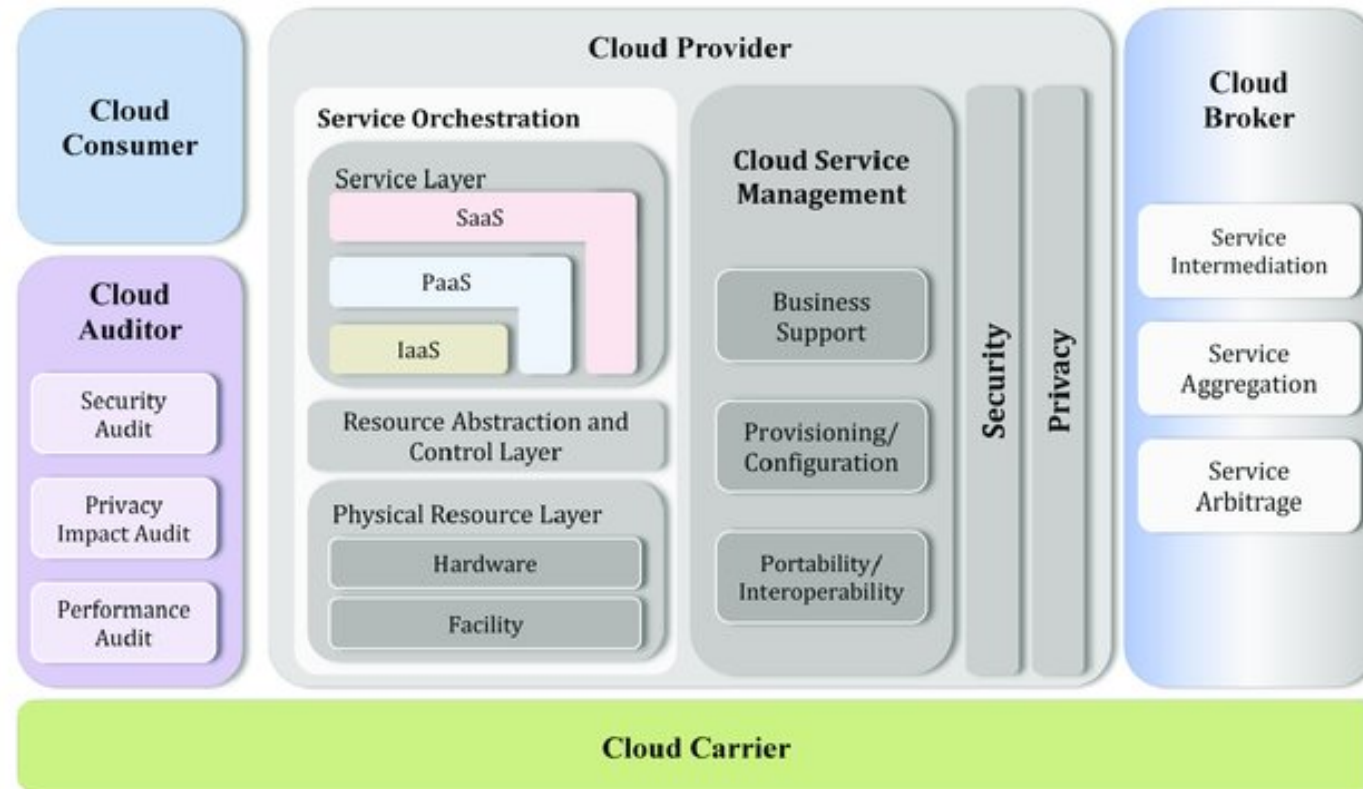
2.1.1 Platform as Service

Platform as a Service (PaaS) provides a runtime environment. It allows programmers to easily create, test, run, and deploy web applications. You can purchase these applications from a cloud service provider on a pay-as-per use basis and access them using the Internet connection. In PaaS, back end scalability is managed by the cloud service provider, so end- users do not need to worry about managing the infrastructure.



PaaS providers provide the Programming languages, Application frameworks, Databases, and Other tools:

- 1. Programming languages :** PaaS providers provide various programming languages for the developers to develop the applications. Some popular programming languages provided by PaaS providers are Java, PHP, Ruby, Perl, and Go.
- 2. Application frameworks :** PaaS providers provide application frameworks to easily understand the application development. Some popular application frameworks provided by PaaS providers are Node.js, Drupal, Joomla, WordPress, Spring, Play, Rack, and Zend.
- 3. Databases :** PaaS providers provide various databases such as ClearDB, PostgreSQL, MongoDB, and Redis to communicate with the applications.
- 4. Other tools :** PaaS providers provide various other tools that are required to develop, test, and deploy the applications.



1. Cloud Consumer

The **Cloud Consumer** is the end-user or organization that utilizes the cloud services provided by a **Cloud Provider**. This entity is the primary beneficiary of cloud computing and relies on it to perform various tasks or run applications without managing the underlying infrastructure.

Responsibilities:

- Uses services like SaaS, PaaS, or IaaS provided by the cloud provider.
- Establishes agreements and SLAs (Service Level Agreements) with the cloud provider.

2. Cloud Provider

The **Cloud Provider** is the entity or organization responsible for delivering cloud services and managing the underlying infrastructure. It offers scalable resources, platforms, and software to cloud consumers, ensuring reliable service delivery and support.

2. Key Components of a Cloud Provider

This component defines how cloud services are managed, coordinated, and delivered to the consumers. It includes different service models:

Service Layer:

- **SaaS (Software as a Service):**
 - Fully managed software solutions provided to end-users over the internet. The consumer uses the software without managing the hardware or underlying systems.
 - **Example: Google Workspace** (Gmail, Google Docs) allows users to access and work on documents online without installing software.
- **PaaS (Platform as a Service):**
 - Provides platforms for developers to build, deploy, and manage applications without worrying about the underlying infrastructure.
 - **Example: Microsoft Azure App Services** allows developers to deploy applications while Azure handles scalability, load balancing, and server maintenance.
- **IaaS (Infrastructure as a Service):**
 - Virtualized resources like servers, storage, and networking that allow consumers to build their own IT environment.
 - **Example: Amazon EC2 (Elastic Compute Cloud)** provides virtual machines for hosting applications, with flexibility to scale up or down as needed.

B. Resource Abstraction and Control Layer

- Abstracts and manages physical resources such as servers, storage, and network devices.
- Provides tools to provision and control these resources, ensuring flexibility and efficiency.
- **Example: AWS Elastic Load Balancer (ELB)** abstracts physical load balancers and distributes incoming application traffic across multiple virtual servers.

C. Physical Resource Layer

- This includes:
 - **Hardware:** Physical servers, storage devices, and networking components.
 - **Facilities:** Data centers housing the hardware, ensuring proper environmental control (e.g., cooling and power supply).
- **Example: Google's Data Centers** are equipped with advanced cooling technologies and high-performance servers to provide seamless cloud services.

D. Cloud Service Management

- **Business Support:**
 - Handles customer-facing operations like billing, metering, subscription plans, and payment processing.
 - **Example:** AWS offers a **pay-as-you-go** model, where customers are billed based on actual usage.
- **Provisioning/Configuration:**
 - Automates the setup and configuration of cloud services for consumers.
 - **Example:** When a user creates a virtual machine in **Azure**, the provisioning system allocates the necessary resources automatically.
- **Portability/Interoperability:**
 - Ensures cloud services are compatible with other platforms and systems, allowing users to migrate their data or applications across providers seamlessly.
 - **Example:** Using **Kubernetes** for containerized applications enables portability between **AWS, Azure, and Google Cloud**.

E. Security & Privacy

- Provides robust mechanisms for:
 - Data protection
 - Compliance with regulatory standards
 - Authentication and authorization
- **Example: AWS Identity and Access Management (IAM)** ensures secure access control for resources and applications.

3. Cloud Broker

The **Cloud Broker** serves as an intermediary between cloud providers and cloud consumers. Its primary function is to simplify the process of selecting, integrating, and optimizing cloud services for the consumer. By acting as a middleman, the broker ensures that consumers receive tailored solutions, often combining multiple services into one seamless offering.

2. Key Functions of a Cloud Broker

A. Service Intermediation

The broker **enhances** or **modifies** existing cloud services provided by the cloud provider to meet specific consumer requirements.

This may include adding custom features, improving performance, or optimizing usage.

Example:

A **Cloud Broker** customizes a **SaaS Customer Relationship Management (CRM)** tool by integrating advanced analytics and reporting features for a sales team.

B. Service Aggregation

- The broker combines multiple services from different providers into a **single unified offering**, ensuring they work together seamlessly.
- This is particularly useful when a consumer needs a combination of services from different cloud providers.
- **Example:**
 - A **Cloud Broker** combines:
 - **AWS for infrastructure (IaaS)**
 - **Google Cloud's AI tools (PaaS)**
 - **Microsoft Teams (SaaS)** for collaboration
 - The broker integrates these services into a unified solution for a company's IT needs.

C. Service Arbitrage

- The broker evaluates multiple cloud providers and selects the **best provider or service** based on criteria such as cost, performance, and reliability.
- This ensures the consumer gets the most efficient and cost-effective service.
- **Example:**
 - A **Cloud Broker** compares pricing and latency between **AWS**, **Google Cloud**, and **Azure** for hosting a web application, choosing the provider that offers the lowest cost with the highest performance.

4. Cloud Auditor

The **Cloud Auditor** is an independent entity responsible for conducting impartial assessments of the cloud environment. The auditor ensures that the cloud services meet security, privacy, and performance standards, helping consumers and providers maintain compliance and trustworthiness.

Responsibilities:

A. Security Audit

Evaluates the **security measures** in place to ensure they meet industry standards and effectively protect sensitive data from breaches and threats. Includes verifying compliance with security protocols, encryption standards, and access controls.

B. Privacy Impact Audit

Examines how consumer data is collected, stored, and processed to ensure it complies with **data privacy regulations** like **GDPR**, **CCPA**, or other regional laws. Identifies potential risks related to unauthorized data access or data misuse.

C. Performance Audit

Assesses the **performance** of cloud services, ensuring they meet the agreed-upon **Service Level Agreements (SLAs)** for uptime, latency, scalability, and overall reliability. Ensures that promised performance metrics are achieved consistently.

5. Cloud Carrier

The **Cloud Carrier** serves as the communication backbone that enables the transmission of data and services between **Cloud Providers** and **Cloud Consumers**. It facilitates connectivity and ensures efficient and secure delivery of cloud services.

- Establishes and manages the **network infrastructure** required for data transfer between cloud providers and consumers. Supports protocols like **VPNs, MPLS, or internet services** to ensure seamless connectivity. Eg: An organization accesses a cloud-based CRM system hosted by a provider over a **high-speed internet connection** provided by the Cloud Carrier.
- Implements **encryption, firewalls, and authentication protocols** to safeguard data while it is in transit between the provider and consumer. Prevents unauthorized access and protects against cyber threats like **man-in-the-middle attacks**. Eg: A Cloud Carrier ensures secure communication between a healthcare organization's devices and the cloud by providing **SSL encryption** for all transmitted patient data.
- Ensures **low latency, high bandwidth**, and reliable connections to support real-time applications like video conferencing or IoT systems. Monitors and optimizes the network to handle varying workloads efficiently. Eg: A global video streaming service relies on a Cloud Carrier to deliver uninterrupted, high-quality streaming to users across different regions.

2.1.1 Platform as Service

Advantages :

1) Simplified Development : PaaS allows developers to focus on development and innovation without worrying about infrastructure management.

2) Lower risk

No need for up-front investment in hardware and software. Developers only need a PC and an internet connection to start building applications.

3) Prebuilt business functionality

Some PaaS vendors also provide already defined business functionality so that users can avoid building everything from very scratch and hence can directly start the projects only.

4) Instant community

PaaS vendors frequently provide online communities where the developer can get the ideas to share experiences and seek advice from others.

5) Scalability

Applications deployed can scale from one to thousands of users without any changes to the applications.

2.1.1 Platform as Service

Disadvantages :

1) Vendor lock-in

One has to write the applications according to the platform provided by the PaaS vendor, so the migration of an application to another PaaS vendor would be a problem.

2) Data Privacy

Corporate data, whether it can be critical or not, will be private, so if it is not located within the walls of the company, there can be a risk in terms of privacy of data.

3) Integration with the rest of the systems applications

It may happen that some applications are local, and some are in the cloud. So there will be chances of increased complexity when we want to use data which in the cloud with the local data.

2.1.1 Platform as Service

Popular PaaS providers and services that are provided by them are :

Providers	Services
Google App Engine (GAE)	App Identity, URL Fetch, Cloud storage client library, Logservice
Salesforce.com	Faster implementation, Rapid scalability, CRM Services, Sales cloud, Mobile connectivity, Chatter.
Windows Azure	Compute, security, IoT, Data Storage.
AppFog	Justcloud.com, SkyDrive, GoogleDocs
Openshift	RedHat, Microsoft Azure.
Cloud Foundry from VMware	Data, Messaging, and other services.

2.1 Cloud Reference Model



2.1.2 Software as Service

SaaS is also known as "On-Demand Software". It is a software distribution model in which services are hosted by a cloud service provider. These services are available to end-users over the internet so, the end-users do not need to install any software on their devices to access these services.

There are the following services provided by SaaS providers -

- 1. Business Services** - SaaS Provider provides various business services to start-up the business. The SaaS business services include ERP (Enterprise Resource Planning), CRM (Customer Relationship Management), billing, and sales.
- 2. Document Management** - SaaS document management is a software application offered by a third party (SaaS providers) to create, manage, and track electronic documents.

Example: Slack, Samepage, Box, and Zoho Forms.

- 3. Social Networks** - Social networking sites are used by the general public, so social networking service providers use SaaS for their convenience and handle the general public's information.

Mail Services - To handle the unpredictable number of users and load on e-mail services, many e-mail providers offering their services using SaaS.

2.1.2 Software as Service

Advantages :

1) SaaS is easy to buy

SaaS pricing is based on a monthly fee or annual fee subscription, so it allows organizations to access business functionality at a low cost, which is less than licensed applications.

Unlike traditional software, which is sold as a licensed based with an up-front cost (and often an optional ongoing support fee), SaaS providers are generally pricing the applications using a subscription fee, most commonly a monthly or annually fee.

2. One to Many

SaaS services are offered as a one-to-many model means a single instance of the application is shared by multiple users.

3. Less hardware required for SaaS

The software is hosted remotely, so organizations do not need to invest in additional hardware.

4. Low maintenance required for SaaS

Software as a service removes the need for installation, set-up, and daily maintenance for the organizations. The initial set-up cost for SaaS is typically less than the enterprise software. SaaS vendors are pricing their applications based on some usage parameters, such as a number of users using the application. So SaaS does easy to monitor and automatic updates.

5. No special software or hardware versions required

All users will have the same version of the software and typically access it through the web browser. SaaS reduces IT support costs by outsourcing hardware and software maintenance and support to the IaaS provider.

6. Multidevice support

SaaS services can be accessed from any device such as desktops, laptops, tablets, phones, and thin clients.

7. API Integration

SaaS services easily integrate with other software or services through standard APIs.

8. No client-side installation

SaaS services are accessed directly from the service provider using the internet connection, so do not need to require any software installation.

2.1.2 Software as Service

Disadvantages :

1) Security

Actually, data is stored in the cloud, so security may be an issue for some users. However, cloud computing is not more secure than in-house deployment.

2) Latency issue

Since data and applications are stored in the cloud at a variable distance from the end-user, there is a possibility that there may be greater latency when interacting with the application compared to local deployment. Therefore, the SaaS model is not suitable for applications whose demand response time is in milliseconds.

3) Total Dependency on Internet

Without an internet connection, most SaaS applications are not usable.

4) Switching between SaaS vendors is difficult

Switching SaaS vendors involves the difficult and slow task of transferring the very large data files over the internet and then converting and importing them into another SaaS also.

2.1.3 Software as Service

Popular SaaS providers and services that are provided by them are :

Provider	Services
Salseforce.com	On-demand CRM solutions
Microsoft Office 365	Online office suite
Google Apps	Gmail, Google Calendar, Docs, and sites
NetSuite	ERP, accounting, order management, CRM, Professionals Services Automation (PSA), and e-commerce applications.
GoToMeeting	Online meeting and video-conferencing software
Constant Contact	E-mail marketing, online survey, and event marketing
Oracle CRM	CRM applications
Workday, Inc	Human capital management, payroll, and financial management.

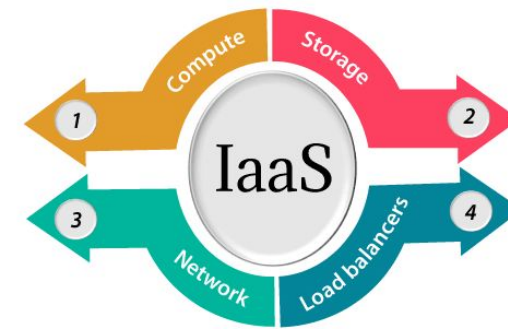
2.1 Cloud Reference Model

2.1.3 Infrastructure as Service

IaaS is also known as Hardware as a Service (HaaS). It is one of the layers of the cloud computing platform. It allows customers to outsource their IT infrastructures such as servers, networking, processing, storage, virtual machines, and other resources. Customers access these resources on the Internet using a pay-as-per use model.

IaaS provider provides the following services -

- 1. Compute:** Computing as a Service includes virtual central processing units and virtual main memory for the Vms that is provisioned to the end- users.
- 2. Storage:** IaaS provider provides back-end storage for storing files.
- 3. Network:** Network as a Service (NaaS) provides networking components such as routers, switches, and bridges for the Vms.
- 4. Load balancers:** It provides load balancing capability at the infrastructure layer.



2.1.3 Infrastructure as Service

Advantages :

1. Shared infrastructure

IaaS allows multiple users to share the same physical infrastructure.

2. Web access to the resources

IaaS allows IT users to access resources over the internet.

3. Pay-as-per-use model

IaaS providers provide services based on the pay-as-per-use basis. The users are required to pay for what they have used.

4. Focus on the core business

IaaS providers focus on the organization's core business rather than on IT infrastructure.

5. On-demand scalability

On-demand scalability is one of the biggest advantages of IaaS. Using IaaS, users do not worry about to upgrade software and troubleshoot the issues related to hardware components.

2.1.3 Infrastructure as Service

Disadvantages :

1. Security

Security is one of the biggest issues in IaaS. Most of the IaaS providers are not able to provide 100% security.

2. Maintenance & Upgrade

Although IaaS service providers maintain the software, but they do not upgrade the software for some organizations.

3. Interoperability issues

It is difficult to migrate VM from one IaaS provider to the other, so the customers might face problem related to vendor lock-in.

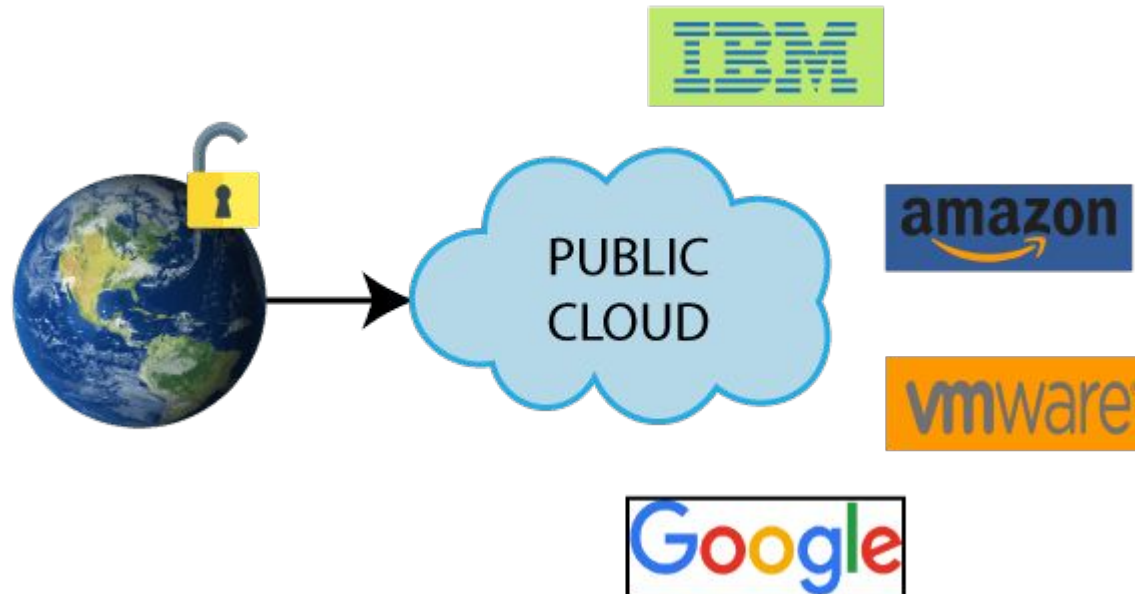
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Windows Azure	Compute, security, IoT, Data Storage.
AppFog	Justcloud.com, SkyDrive, GoogleDocs
Openshift	RedHat, Microsoft Azure.
Cloud Foundry from VMware	Data, Messaging, and other services.

2.2.1 Public Clouds

- Public Cloud provides a shared platform that is accessible to the general public through an Internet connection.
- Public cloud operated on the pay-as-per-use model and administrated by the third party, i.e., Cloud service provider.
- In the Public cloud, the same storage is being used by multiple users at the same time.
- Public cloud is owned, managed, and operated by businesses, universities, government organizations, or a combination of them.
- Amazon Elastic Compute Cloud (EC2), Microsoft Azure, IBM's Blue Cloud, Sun Cloud, and Google Cloud are examples of the public cloud.



2.2.1 Public Clouds

Advantages of Public Cloud

1) Low Cost

Public cloud has a lower cost than private, or hybrid cloud, as it shares the same resources with a large number of consumers.

2) Location Independent

Public cloud is location independent because its services are offered through the internet.

3) Save Time

In Public cloud, the cloud service provider is responsible for the manage and maintain data centers in which data is stored, so the cloud user can save their time to establish connectivity, deploying new products, release product updates, configure, and assemble servers.

4) Quickly and easily set up

Organizations can easily buy public cloud on the internet and deployed and configured it remotely through the cloud service provider within a few hours.

5) Business Agility

Public cloud provides an ability to elastically re-size computer resources based on the organization's requirements.

6) Scalability and reliability

Public cloud offers scalable (easy to add and remove) and reliable (24*7 available) services to the users at an affordable cost.

2.2.1 Public Clouds

Disadvantages of Public Cloud

1) Low Security

Public Cloud is less secure because resources are shared publicly.

2) Performance

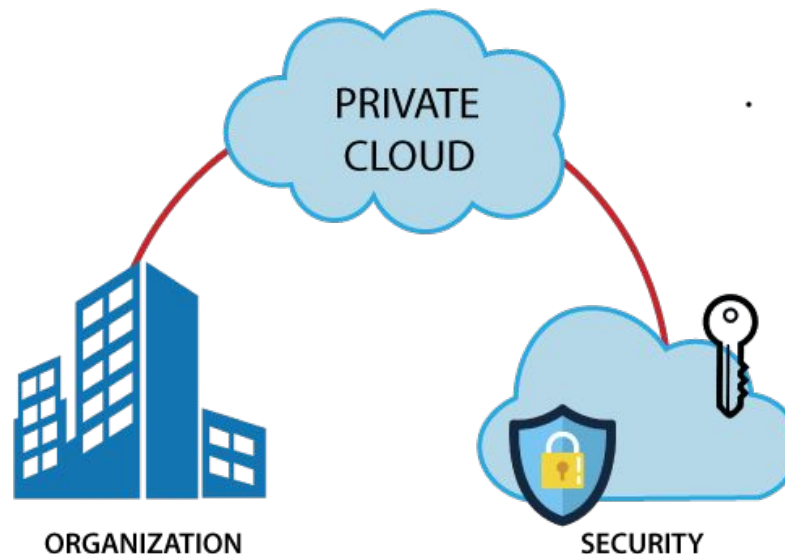
In the public cloud, performance depends upon the speed of internet connectivity.

3) Less customizable

Public cloud is less customizable than the private cloud.

2.2.2 Private Clouds

- Private cloud is also known as an internal cloud or corporate cloud.
- Private cloud provides computing services to a private internal network (within the organization) and selected users instead of the general public.
- Private cloud provides a high level of security and privacy to data through firewalls and internal hosting. It also ensures that operational and sensitive data are not accessible to third-party providers.
- HP Data Centers, Microsoft, Elasta-private cloud, and Ubuntu are the example of a private cloud.



2.2.2 Private Clouds

Advantages

1) More Control

Private clouds have more control over their resources and hardware than public clouds because it is only accessed by selected users.

2) Security & privacy

Security & privacy are one of the big advantages of cloud computing. Private cloud improved the security level as compared to the public cloud.

3) Improved performance

Private cloud offers better performance with improved speed and space capacity.

2.2.2 Private Clouds

Disadvantages

1) High cost

The cost is higher than a public cloud because set up and maintain hardware resources are costly.

2) Restricted area of operations

As we know, private cloud is accessible within the organization, so the area of operations is limited.

3) Limited scalability

Private clouds are scaled only within the capacity of internal hosted resources.

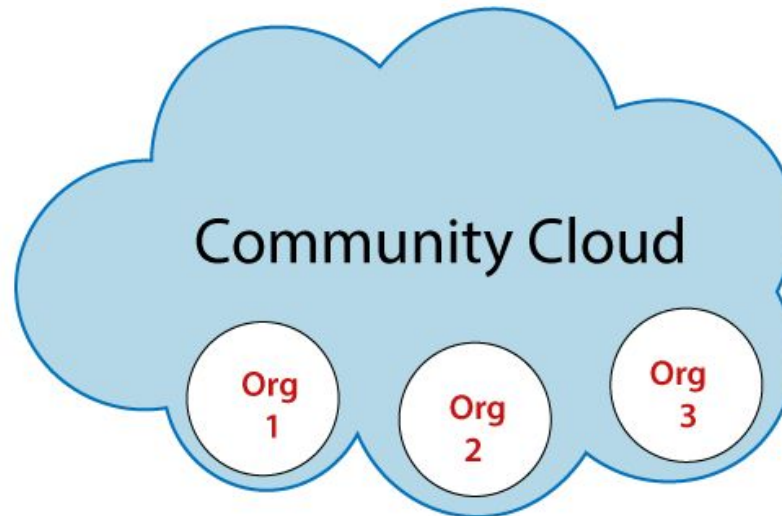
4) Skilled people

Skilled people are required to manage and operate cloud services.

2.2.3 Community Clouds

Community cloud is a cloud infrastructure that allows systems and services to be accessible by a group of several organizations to share the information. It is owned, managed, and operated by one or more organizations in the community, a third party, or a combination of them.

Example: Our government organization within India may share computing infrastructure in the cloud to manage data.



2.2.3 Community Clouds

Advantages

1. Cost effective

Community cloud is cost effective because the whole cloud is shared between several organizations or a community.

2. Flexible and Scalable

The community cloud is flexible and scalable because it is compatible with every user. It allows the users to modify the documents as per their needs and requirement.

3. Security

Community cloud is more secure than the public cloud but less secure than the private cloud.

4. Sharing infrastructure

Community cloud allows us to share cloud resources, infrastructure, and other capabilities among various organizations.

2.2.3 Community Clouds

Disadvantages

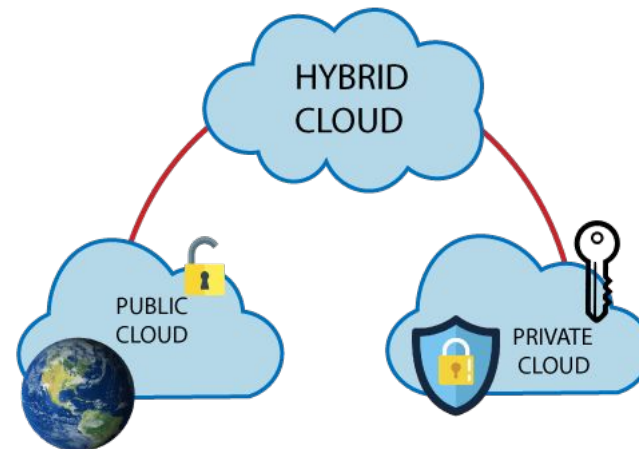
- Community cloud is not a good choice for every organization.
- Slow adoption to data
- The fixed amount of data storage and bandwidth is shared among all community members.
- Community Cloud is costly than the public cloud.
- Sharing responsibilities among organizations is difficult.

2.2.4 Hybrid Clouds

- Hybrid cloud is a combination of public and private clouds.

Hybrid cloud = public cloud + private cloud

- The main aim to combine these cloud (Public and Private) is to create a unified, automated, and well-managed computing environment.
- In the Hybrid cloud, non-critical activities are performed by the public cloud and critical activities are performed by the private cloud.
- Mainly, a hybrid cloud is used in finance, healthcare, and Universities.
- The best hybrid cloud provider companies are Amazon, Microsoft, Google, Cisco, and NetApp.



2.2.4 Hybrid Clouds

Advantages

1) Flexible and secure

- It provides flexible resources because of the public cloud and secure resources because of the private cloud.

2) Cost effective

- Hybrid cloud costs less than the private cloud. It helps organizations to save costs for both infrastructure and application support.

3) Cost effective

- It offers the features of both the public as well as the private cloud. A hybrid cloud is capable of adapting to the demands that each company needs for space, memory, and system.

4) Security

- Hybrid cloud is secure because critical activities are performed by the private cloud.

5) Risk Management

- Hybrid cloud provides an excellent way for companies to manage the risk.

2.2.4 Hybrid Clouds

Disadvantages

1) Networking issues

In the Hybrid Cloud, networking becomes complex because of the private and the public cloud.

2) Infrastructure Compatibility

Infrastructure compatibility is the major issue in a hybrid cloud. With dual-levels of infrastructure, a private cloud controls the company, and a public cloud does not, so there is a possibility that they are running in separate stacks.

3) Reliability

The reliability of the services depends on cloud service providers.

2.3 Cloud design and implementation using SOA

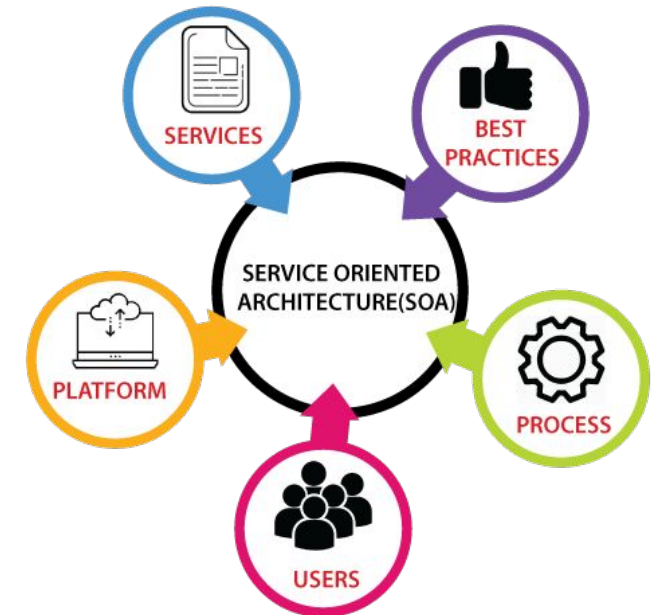
Service-Oriented Architecture (SOA) allows organizations to access on-demand cloud-based computing solutions according to the change of business needs. It can work without or with cloud computing. The advantages of using SOA is that it is easy to maintain, platform independent, and highly scalable.

Service Provider and Service consumer are the two major roles within SOA.

Applications of Service-Oriented Architecture

There are the following applications of Service-Oriented Architecture -

1. It is used in the healthcare industry.
2. It is used to create many mobile applications and games.
3. In the air force, SOA infrastructure is used to deploy situational awareness systems.
4. The service-oriented architecture is shown below:



2.4 Security, Trust and Privacy

Cloud computing provides various advantages, such as improved collaboration, excellent accessibility, Mobility, Storage capacity, etc. But there are also security risks in cloud computing.

1. Data Loss
2. Hacked Interfaces and Insecure APIs
3. Data Breach
4. Vendor lock-in
5. Increased complexity and trained IT staff
6. Denial of Service (DoS) attacks
7. Account hijacking